Proof for Theorem 4 2 (Strong Convexity With Respect To Beta-Star; Thm:global-Strong-Convexity-Main)

Generated by Statopia Prover

October 29, 2026

Key Lemmas

Lemma 1 (Coordinate directional derivative of the population squared error)**.** *Let $p \in \mathbb{N}$, let μ be a probability measure on $\mathbb{R}^p \times \mathbb{R}$, let $\beta \in \mathbb{R}^p$, and let $j \in \{1, \dots, p\}$. For $v, x \in \mathbb{R}^p$, write*

$$v \cdot x = \sum_{i=1}^{p} v_i x_i,$$

and for $z = (x, y) \in \mathbb{R}^p \times \mathbb{R}$ define the population squared error functional

$$R_\mu(v) = \int_{\mathbb{R}^p \times \mathbb{R}} \big(y - v \cdot x\big)^2 \, d\mu(x, y).$$

Let e_j be the j-th standard basis vector of $\mathbb{R}^p$, and define the coordinate directional derivative

$$D_j R_\mu(v) = \left. \frac{d}{dt} \right|_{t=0} R_\mu(v + t e_j).$$

Assume:

(i) $z = (x, y) \mapsto (y - \beta \cdot x)^2$ *is integrable with respect to μ;*

(ii) $z = (x, y) \mapsto (y - \beta \cdot x) x_j$ *is integrable with respect to μ;*

(iii) $z = (x, y) \mapsto x_j^2$ *is integrable with respect to μ.*

Then

$$D_j R_\mu(\beta) = -2 \int_{\mathbb{R}^p \times \mathbb{R}} (y - \beta \cdot x) x_j \, d\mu(x, y).$$

Proof. Write $z = (x, y) \in \mathbb{R}^p \times \mathbb{R}$, and put

$$r_\beta(z) = y - \beta \cdot x.$$

For every real t, first compute the shifted linear form. The i-th coordinate of $\beta + te_j$ is

$$\beta_i + t\,\mathbf{1}_{\{i=j\}}.$$

Hence

$$(\beta + te_j) \cdot x = \sum_{i=1}^{p} \big(\beta_i + t\,\mathbf{1}_{\{i=j\}}\big) x_i = \sum_{i=1}^{p} \beta_i x_i + t \sum_{i=1}^{p} \mathbf{1}_{\{i=j\}} x_i = \beta \cdot x + t x_j.$$

Consequently, for every t and every $z = (x, y)$,

$$\begin{aligned} \big(y - (\beta + te_j) \cdot x\big)^2 &= \big(y - \beta \cdot x - t x_j\big)^2 \\ &= (y - \beta \cdot x)^2 - 2t\,(y - \beta \cdot x) x_j + t^2 x_j^2 \\ &= r_\beta(z)^2 - 2t\, r_\beta(z) x_j + t^2 x_j^2. \end{aligned}$$

Now set

$$A = \int_{\mathbb{R}^p \times \mathbb{R}} r_\beta(z)^2 \, d\mu(z), \qquad B = \int_{\mathbb{R}^p \times \mathbb{R}} r_\beta(z) x_j \, d\mu(z), \qquad C = \int_{\mathbb{R}^p \times \mathbb{R}} x_j^2 \, d\mu(z).$$

These quantities are finite by Conditions (i)–(iii). Moreover, for each fixed t, the function

$$z \mapsto 2t\, r_\beta(z) x_j$$

is integrable as a scalar multiple of the function in Condition (ii), and the function

$$z \mapsto t^2 x_j^2$$

is integrable as a scalar multiple of the function in Condition (iii). The difference

$$z \mapsto r_\beta(z)^2 - 2t\, r_\beta(z) x_j$$

is therefore integrable by Condition (i), so linearity of the integral applies to the three summands above. Using the pointwise identity and then homogeneity and additivity of the

integral, we obtain for every real t

$$\begin{aligned} R_\mu(\beta + te_j) &= \int_{\mathbb{R}^p \times \mathbb{R}} \left(r_\beta(z)^2 - 2t\, r_\beta(z) x_j + t^2 x_j^2 \right) d\mu(z) \\ &= A - 2tB + t^2 C. \end{aligned}$$

Define the quadratic function

$$F(t) = A - 2tB + t^2 C.$$

The first term is constant and has derivative 0. The second term is linear and has derivative $-2B$. The third term is quadratic and has derivative $2tC$ at t, hence derivative 0 at $t = 0$. Therefore

$$F'(0) = 0 - 2B + 0 = -2B.$$

Since $R_\mu(\beta + te_j) = F(t)$ for every real t, the coordinate directional derivative is

$$D_j R_\mu(\beta) = \left. \frac{d}{dt} \right|_{t=0} R_\mu(\beta + te_j) = F'(0) = -2B.$$

Finally, substituting the definition of B gives

$$D_j R_\mu(\beta) = -2 \int_{\mathbb{R}^p \times \mathbb{R}} (y - \beta \cdot x) x_j \, d\mu(x, y).$$

This proves the asserted identity.

□

Lemma 2. *Let p be a natural number and let I be a finite nonempty index set. For each $e \in I$, let μ_e be a probability measure on $\mathbb{R}^p \times \mathbb{R}$. Write $z = (x, y)$, $\langle b, x \rangle = \sum_{i=1}^p b_i x_i$, and $\operatorname{supp} b = \{j : b_j \neq 0\}$. Define*

$$R_e(b) = \int_{\mathbb{R}^p \times \mathbb{R}} \left(y - \langle b, x \rangle \right)^2 d\mu_e(x, y).$$

Let C_e be the covariance matrix with entries

$$(C_e)_{ij} = \int_{\mathbb{R}^p \times \mathbb{R}} x_i x_j \, d\mu_e(x, y),$$

and write $v^\top C_e v = \sum_{i,j} (C_e)_{ij} v_i v_j$. Let $\beta^\star \in \mathbb{R}^p$ and $S^\star \subseteq \{1, \ldots, p\}$ satisfy

$$j \in S^\star \iff \beta_j^\star \neq 0.$$

Let $\beta \in \mathbb{R}^p$. For every $e \in I$ and every finite $S \subseteq \{1, \ldots, p\}$, let $\beta^{(e,S)} \in \mathbb{R}^p$. Assume the following conditions:

(i) For all $e \in I$ and finite $S \subseteq \{1, \ldots, p\}$,

$$\operatorname{supp} \beta^{(e,S)} \subseteq S,$$

and $\beta^{(e,S)}$ minimizes R_e over the set

$$\{b \in \mathbb{R}^p : \operatorname{supp} b \subseteq S\}.$$

(ii) There is $\kappa_L > 0$ such that for all $e \in I$ and $v \in \mathbb{R}^p$,

$$\kappa_L \|v\|^2 \leq v^\top C_e v.$$

(iii) There are weights ω_e satisfying

$$\omega_e = \frac{1}{|I|} \qquad (e \in I).$$

(iv) For all $e \in I$, $b \in \mathbb{R}^p$, and $j \in \{1, \ldots, p\}$, the function

$$(x, y) \mapsto \big(y - \langle b, x \rangle\big) x_j$$

is μ_e-integrable.

(v) For all $e \in I$, $b \in \mathbb{R}^p$, and $j \in \{1, \ldots, p\}$, the coordinate partial derivative

$$D_e(b, j) = \frac{\partial}{\partial t} R_e(b + t e_j)\bigg|_{t=0}$$

exists and satisfies

$$D_e(b, j) = -2 \int_{\mathbb{R}^p \times \mathbb{R}} \big(y - \langle b, x \rangle\big) x_j \, d\mu_e(x, y).$$

Define

$$J(\beta) = \sum_{j \in \operatorname{supp} \beta} \sum_{e \in I} \frac{\omega_e}{4} D_e(\beta, j)^2,$$

and, for finite $S \subseteq \{1, \dots, p\}$,

$$\big(\bar{\beta}^{(S)}\big)_j = \frac{1}{|I|} \sum_{e \in I} \beta_j^{(e,S)}, \qquad \Delta(S) = \sum_{e \in I} \frac{1}{|I|} \big\| \beta^{(e,S)} - \bar{\beta}^{(S)} \big\|^2.$$

Then

$$J(\beta) \geq \kappa_L^2 \, \Delta(\operatorname{supp} \beta).$$

Proof. Put $S = \operatorname{supp} \beta$. We use the standard coordinate identities

$$\|x\|^2 = \sum_{i=1}^{p} x_i^2, \qquad \langle e_i, x \rangle = x_i, \qquad \langle c e_i, x \rangle = c x_i,$$

where e_i is the i-th standard basis vector. We also use that vector addition, subtraction, and scalar multiplication are coordinatewise, and that for finite sums,

$$\left(\sum_{j \in T} f_j g_j \right)^2 \leq \left(\sum_{j \in T} f_j^2 \right) \left(\sum_{j \in T} g_j^2 \right).$$

Moreover, if $S \subseteq T$ and the omitted terms are nonnegative, then the sum over S is at most the sum over T, with equality when the omitted terms vanish.

First we record an integrability fact. For every $e \in I$ and every $i, j \in \{1, \dots, p\}$, the function $x_i x_j$ is μ_e-integrable. Indeed, applying condition (iv) with $b = e_i$ and with $b = 2e_i$, both $(y - x_i)x_j$ and $(y - 2x_i)x_j$ are μ_e-integrable; their difference is

$$(y - x_i)x_j - (y - 2x_i)x_j = x_i x_j.$$

Thus differences of integrable functions give the required integrability.

Let

$$w_j^{(e)} = \int_{\mathbb{R}^p \times \mathbb{R}} \big(y - \langle \beta, x \rangle\big) x_j \, d\mu_e(x, y).$$

For fixed $e \in I$ and $j \in S$, consider the line

$$t \mapsto \beta^{(e,S)} + t e_j.$$

For every t, this vector remains supported in S: if $i \notin S$, then $i \neq j$, and condition (i) gives $\beta_i^{(e,S)} = 0$, so the i-th coordinate of $\beta^{(e,S)} + t e_j$ is zero. Hence this line lies in the constraint set on which $\beta^{(e,S)}$ is a minimizer. By the first-order condition for a constrained minimizer

on this linear constraint set (Lemma 1), the derivative at $t = 0$ vanishes:

$$D_e(\beta^{(e,S)}, j) = 0.$$

Using condition (v), we obtain

$$\int_{\mathbb{R}^p \times \mathbb{R}} \big(y - \langle \beta^{(e,S)}, x \rangle\big) x_j \, d\mu_e(x, y) = 0 \qquad (j \in S).$$

Next, for fixed $e \in I$ and $j \in S$, decompose

$$y - \langle \beta, x \rangle = \big(y - \langle \beta^{(e,S)}, x \rangle\big) + \big\langle \beta^{(e,S)} - \beta, x \big\rangle.$$

Multiplying by x_j and integrating, the first term has integral 0 by the first-order condition above. Therefore, using the integrability of the cross terms and of $x_i x_j$,

$$\begin{aligned} w_j^{(e)} &= \int_{\mathbb{R}^p \times \mathbb{R}} \big(y - \langle \beta, x \rangle\big) x_j \, d\mu_e(x, y) \\ &= \sum_{i=1}^{p} \big(\beta_i^{(e,S)} - \beta_i\big) \int_{\mathbb{R}^p \times \mathbb{R}} x_i x_j \, d\mu_e(x, y) \\ &= \sum_{i=1}^{p} \big(\beta_i^{(e,S)} - \beta_i\big)(C_e)_{ij}. \end{aligned}$$

Now fix $e \in I$ and set

$$v_e = \beta^{(e,S)} - \beta.$$

Condition (i) gives $\operatorname{supp} \beta^{(e,S)} \subseteq S$, and $S = \operatorname{supp} \beta$, so v_e is supported in S. In particular, $(v_e)_i = 0$ for $i \notin S$. Hence

$$\sum_{j \in S} (v_e)_j w_j^{(e)} = \sum_{j \in S} (v_e)_j \sum_{i=1}^{p} (v_e)_i (C_e)_{ij} = \sum_{i,j=1}^{p} (C_e)_{ij} (v_e)_i (v_e)_j = v_e^\top C_e v_e.$$

By condition (ii),

$$v_e^\top C_e v_e \geq \kappa_L \|v_e\|^2.$$

On the other hand, the Cauchy–Schwarz inequality for finite sums gives

$$\left(\sum_{j \in S} (v_e)_j w_j^{(e)} \right)^2 \leq \left(\sum_{j \in S} (v_e)_j^2 \right) \left(\sum_{j \in S} (w_j^{(e)})^2 \right) \leq \|v_e\|^2 \sum_{j \in S} (w_j^{(e)})^2.$$

If $v_e = 0$, the desired estimate is immediate because the left-hand side is 0 and the right-hand side is a sum of squares. If $v_e \neq 0$, then combining the lower bound with the Cauchy–Schwarz estimate yields

$$\kappa_L^2 \|v_e\|^4 \leq \left(\sum_{j \in S} (v_e)_j w_j^{(e)} \right)^2 \leq \|v_e\|^2 \sum_{j \in S} (w_j^{(e)})^2.$$

Dividing by $\|v_e\|^2 > 0$, we obtain

$$\kappa_L^2 \|\beta^{(e,S)} - \beta\|^2 \leq \sum_{j \in S} (w_j^{(e)})^2.$$

We now rewrite $J(\beta)$. By its definition and condition (v),

$$D_e(\beta, j) = -2 w_j^{(e)},$$

so

$$\begin{aligned} J(\beta) &= \sum_{j \in S} \sum_{e \in I} \frac{\omega_e}{4} D_e(\beta, j)^2 \\ &= \sum_{j \in S} \sum_{e \in I} \omega_e \, (w_j^{(e)})^2 \\ &= \sum_{e \in I} \omega_e \sum_{j \in S} (w_j^{(e)})^2. \end{aligned}$$

Using the estimate from the preceding paragraph and $\omega_e \geq 0$,

$$\sum_{e \in I} \omega_e \, \kappa_L^2 \|\beta^{(e,S)} - \beta\|^2 \leq \sum_{e \in I} \omega_e \sum_{j \in S} (w_j^{(e)})^2 = J(\beta).$$

Since $\omega_e = |I|^{-1}$, this says

$$\kappa_L^2 \sum_{e \in I} \frac{1}{|I|} \|\beta^{(e,S)} - \beta\|^2 \leq J(\beta).$$

It remains to compare the last left-hand side with $\Delta(S)$. Let

$$a_e = \beta^{(e,S)}, \qquad \bar{a} = \bar{\beta}^{(S)}.$$

By definition of $\bar{a}$, for every coordinate j,

$$\sum_{e \in I} \omega_e (a_e - \bar{a})_j = 0,$$

and since $\omega_e = |I|^{-1}$, also $\sum_{e\in I}\omega_e = 1$. Expanding the squared Euclidean norm gives, for each e,

$$\|a_e - \beta\|^2 = \|a_e - \bar{a}\|^2 + 2(a_e - \bar{a})\cdot(\bar{a} - \beta) + \|\bar{a} - \beta\|^2.$$

Taking the weighted sum over $e \in I$, the cross term vanishes because

$$\sum_{e\in I}\omega_e(a_e - \bar{a}) = 0$$

coordinatewise, while

$$\sum_{e\in I}\omega_e\|\bar{a} - \beta\|^2 = \|\bar{a} - \beta\|^2.$$

Therefore

$$\sum_{e\in I}\omega_e\|a_e - \beta\|^2 = \sum_{e\in I}\omega_e\|a_e - \bar{a}\|^2 + \|\bar{a} - \beta\|^2 \geq \sum_{e\in I}\omega_e\|a_e - \bar{a}\|^2.$$

The right-hand side is exactly $\Delta(S)$. Since $\omega_e = |I|^{-1}$, we have

$$\Delta(S) \leq \sum_{e\in I}\frac{1}{|I|}\|\beta^{(e,S)} - \beta\|^2.$$

Combining this with the previous estimate and using $\kappa_L^2 \geq 0$, we conclude

$$\kappa_L^2\Delta(S) \leq \kappa_L^2\sum_{e\in I}\frac{1}{|I|}\|\beta^{(e,S)} - \beta\|^2 \leq J(\beta).$$

Since $S = \operatorname{supp}\beta$, this is

$$J(\beta) \geq \kappa_L^2\Delta(\operatorname{supp}\beta),$$

as required.

□

Lemma 3. *Let $p \in \mathbb{N}$, let I be a finite nonempty set of environments, and write $[p] = \{1,\dots,p\}$. For each $e \in I$, let μ_e be a Borel measure on $\mathbb{R}^p \times \mathbb{R}$, whose elements are written $z = (x, y)$. For a finite $S \subseteq [p]$, let $\mathcal{F}_S$ be the sub-σ-algebra of the Borel σ-algebra on $\mathbb{R}^p \times \mathbb{R}$ generated by the coordinate functions x_j with $j \in S$. Put*

$$\rho_{\beta^*}(z) = y - \langle\beta^*, x\rangle,$$

and, for $j \in [p]$,

$$\bar{u}_j = \frac{1}{|I|} \sum_{e \in I} \int \rho_{\beta^*}(z) x_j \, d\mu_e(z), \qquad b(S) = \sum_{j \in S} \bar{u}_j^2.$$

Assume:

(i) each μ_e *is a probability measure;*

(ii) $S^* \subseteq [p]$ *is exactly the support of* β^**, that is,*

$$j \in S^* \iff \beta_j^* \neq 0;$$

(iii) for every $e \in I$*, the function* $z \mapsto y$ *is* μ_e*-integrable;*

(iv) for every $e \in I$*,*

$$\mathbb{E}_{\mu_e}[y \mid \mathcal{F}_{S^*}](z) = \langle \beta^*, x \rangle \qquad \mu_e\text{-a.e.};$$

(v) for every $e \in I$ *and* $j \in [p]$*, the function*

$$z \mapsto \rho_{\beta^*}(z) x_j$$

is μ_e*-integrable.*

Then, for every $e \in I$ *and* $j \in S^*$*,*

$$\int \rho_{\beta^*}(z) x_j \, d\mu_e(z) = 0;$$

for every $j \in S^*$*,*

$$\frac{1}{|I|} \sum_{e \in I} \int \rho_{\beta^*}(z) x_j \, d\mu_e(z) = 0;$$

and for every finite $S \subseteq [p]$*,*

$$b(S) = \sum_{j \in S \cup S^*} \bar{u}_j^2.$$

Proof. Write $\mathcal{F}_* = \mathcal{F}_{S^*}$ and $\rho(z) = \rho_{\beta^*}(z) = y - \langle \beta^*, x \rangle$. We first prove the following claim: for every $e \in I$ and every $j \in S^*$,

$$\int \rho(z) x_j \, d\mu_e(z) = 0.$$

Fix $e \in I$ and $j \in S^*$. Since μ_e is a probability measure, its restriction to the sub-σ-algebra $\mathcal{F}_*$ is σ-finite, so conditional expectations with respect to $\mathcal{F}_*$ are well-defined for

integrable functions. The coordinate map

$$\Phi(x, y) = (x_k)_{k \in S^*}$$

is measurable, hence $\mathcal{F}_*$ is a sub-σ-algebra of the Borel σ-algebra on $\mathbb{R}^p \times \mathbb{R}$.

By Condition (iii), y is μ_e-integrable, so its conditional expectation with respect to $\mathcal{F}_*$ is integrable. By Condition (iv), this conditional expectation agrees μ_e-almost everywhere with $\langle \beta^*, x \rangle$. Therefore $\langle \beta^*, x \rangle$ is μ_e-integrable. Consequently

$$\rho(z) = y - \langle \beta^*, x \rangle$$

is μ_e-integrable as the difference of two μ_e-integrable functions.

For every $k \in S^*$, the coordinate x_k is $\mathcal{F}_*$-measurable, because it is a projection of the measurable map Φ. Since S^* is exactly the support of β^*, we have

$$\langle \beta^*, x \rangle = \sum_{k \in S^*} \beta_k^* x_k.$$

Thus $\langle \beta^*, x \rangle$ is $\mathcal{F}_*$-measurable as a finite sum of $\mathcal{F}_*$-measurable functions. It is also μ_e-integrable, so its conditional expectation is itself:

$$\mathbb{E}_{\mu_e}[\langle \beta^*, x \rangle \mid \mathcal{F}_*](z) = \langle \beta^*, x \rangle \qquad \mu_e\text{-a.e.}$$

Using linearity of conditional expectation together with Condition (iv), we obtain

$$\mathbb{E}_{\mu_e}[\rho \mid \mathcal{F}_*](z) = \mathbb{E}_{\mu_e}[y \mid \mathcal{F}_*](z) - \mathbb{E}_{\mu_e}[\langle \beta^*, x \rangle \mid \mathcal{F}_*](z) = 0 \qquad \mu_e\text{-a.e.}$$

Now $j \in S^*$, so x_j is $\mathcal{F}_*$-measurable. Condition (v) says that $\rho(z)x_j$ is μ_e-integrable, and ρ itself is μ_e-integrable. By the pull-out property for conditional expectations,

$$\mathbb{E}_{\mu_e}[\rho x_j \mid \mathcal{F}_*](z) = x_j \, \mathbb{E}_{\mu_e}[\rho \mid \mathcal{F}_*](z) \qquad \mu_e\text{-a.e.},$$

and the same identity holds with the product written in the opposite order by commutativity of multiplication. Hence

$$\begin{aligned} \int \rho(z) x_j \, d\mu_e(z) &= \int \mathbb{E}_{\mu_e}[\rho x_j \mid \mathcal{F}_*](z) \, d\mu_e(z) \\ &= \int x_j \, \mathbb{E}_{\mu_e}[\rho \mid \mathcal{F}_*](z) \, d\mu_e(z) = 0, \end{aligned}$$

because $\mathbb{E}_{\mu_e}[\rho \mid \mathcal{F}_*] = 0$ μ_e-almost everywhere. This proves the claim.

The first conclusion is precisely the claim.

For the second conclusion, fix $j \in S^*$. By the claim, each integral

$$\int \rho(z) x_j \, d\mu_e(z)$$

vanishes. Therefore the finite sum over $e \in I$ is zero, and multiplying by $1/|I|$ gives

$$\frac{1}{|I|} \sum_{e \in I} \int \rho(z) x_j \, d\mu_e(z) = 0.$$

For the third conclusion, fix a finite $S \subseteq [p]$. If

$$j \in (S \cup S^*) \setminus S,$$

then $j \in S^*$. By the claim,

$$\int \rho(z) x_j \, d\mu_e(z) = 0$$

for every $e \in I$. Hence

$$\bar{u}_j = \frac{1}{|I|} \sum_{e \in I} \int \rho(z) x_j \, d\mu_e(z) = 0,$$

so the squared term $\bar{u}_j^2$ vanishes. Since $S \subseteq S \cup S^*$, adding these zero terms to the finite sum defining $b(S)$ yields

$$b(S) = \sum_{j \in S} \bar{u}_j^2 = \sum_{j \in S \cup S^*} \bar{u}_j^2.$$

This proves the third conclusion.

□

External node: cited lemma (AXIOM)

Lemma 4 (Euclidean Cauchy–Schwarz and the two-term quadratic bound)**.** *Let k be a natural number. For all $x, y \in \mathbb{R}^k$,*

$$|\langle x, y \rangle| \le \|x\|_2 \|y\|_2, \qquad \langle x, y \rangle^2 \le \|x\|_2^2 \|y\|_2^2,$$

and for every $t > 0$,

$$2\langle x, y \rangle \le t\|x\|_2^2 + t^{-1}\|y\|_2^2.$$

This lemma is supplied as an external assumption; its proof is not part of the local Lean proof.

External assumption linked to task paper_2c08ddb98e7c4dc4993a0b23e7a07337: retained as external. No completed proof for this consumer is certified by this snapshot.

Source The cited result is the standard Euclidean Cauchy–Schwarz inequality, together with its squared form and the two-term consequence commonly called Young's inequality. No unique bibliographic source, author, or theorem number was supplied; the reference identifies it only as an external standard result.

Reference provenance Here \(\langle x,y\rangle=x^\top y\), and \(\|x\|_2\) is the Euclidean \(\ell_2\)-norm. The proof applies the two-term consequence to \[x=\sqrt{\epsilon}\,\Sigma_{\bar S}^{1/2}\Delta_{\bar S}, \qquad y=\epsilon^{-1/2}\Sigma_{\bar S}^{-1/2}\bar u_{\bar S}, \] where \(\Sigma_{\bar S}\) is positive definite under Condition 1, so \(\Sigma_{\bar S}^{-1/2}\) is well defined. The separate fact \(\bar u_T=0\) is derived from the model/conditional-expectation structure and is not part of CauchySchwarz. No unique bibliographic source pair is supplied for this external standard result. Missing material: None for the mathematical statement; only bibliographic citation metadata is absent.

Use in this proof It is used to bound the cross term in the linear part of the risk comparison. In that application the two-term consequence is applied with a positive parameter $t = \varepsilon$ in $(0,1)$, after rescaling the vectors by $\sqrt{\varepsilon}\,\Sigma_{\bar S}^{1/2}\Delta_{\bar S}$ and $\varepsilon^{-1/2}\Sigma_{\bar S}^{-1/2}\bar u_{\bar S}$, where $\Sigma_{\bar S}$ is positive definite under the model assumptions. This yields the corresponding estimate for the inner product. The separate fact $\bar u_T = 0$ is not part of this external inequality; it comes from the conditional-expectation structure.

Lemma 5 (Strong-convexity estimate for the pooled covariance)**.** *Let $p \geq 0$, and let I be a finite nonempty index set. For each $e \in I$, let μ_e be a measure on $\mathbb{R}^p \times \mathbb{R}$. Write a point of $\mathbb{R}^p \times \mathbb{R}$ as $z = (x, y)$, and put*

$$r_{\beta^*}(x, y) = y - \langle \beta^*, x \rangle.$$

Let $\beta^*, \beta \in \mathbb{R}^p$, let $S^* \subseteq \{1, \ldots, p\}$, and define

$$\operatorname{supp}(\beta) = \{j : \beta_j \neq 0\}, \qquad u_j^{(e)} = \int_{\mathbb{R}^p \times \mathbb{R}} r_{\beta^*}(x, y)\, x_j \, d\mu_e(x, y), \qquad \bar u_j = \frac{1}{|I|} \sum_{e \in I} u_j^{(e)}.$$

Let $\bar{u} = (\bar{u}_1, \dots, \bar{u}_p)$. Define

$$\Sigma_e = \int_{\mathbb{R}^p \times \mathbb{R}} x x^{\mathsf{T}} \, d\mu_e(x, y), \qquad \Sigma = \frac{1}{|I|} \sum_{e \in I} \Sigma_e,$$

and for a vector $v \in \mathbb{R}^p$, write $v^{\mathsf{T}} \Sigma v$ for the associated quadratic form. Assume:

(i) $S^* = \operatorname{supp}(\beta^*)$;

(ii) each μ_e is a probability measure;

(iii) for every $e \in I$ and every $j \in \{1, \dots, p\}$, the function

$$(x, y) \longmapsto r_{\beta^*}(x, y)\, x_j$$

is μ_e-integrable;

(iv) for every $j \notin \operatorname{supp}(\beta) \cup S^*$, one has $(\beta - \beta^*)_j = 0$;

(v) $\bar{u}_j = 0$ for every $j \in S^*$;

(vi) $0 < \kappa_L$;

(vii) $\kappa_L \|v\|^2 \le v^{\mathsf{T}} \Sigma v$ for every $v \in \mathbb{R}^p$.

Then for every $\varepsilon > 0$,

$$2\langle \beta - \beta^*, \bar{u} \rangle \le \varepsilon \, (\beta - \beta^*)^{\mathsf{T}} \Sigma (\beta - \beta^*) + \varepsilon^{-1} \kappa_L^{-1} \sum_{j \in \operatorname{supp}(\beta)} \bar{u}_j^2.$$

Proof. Put

$$\Delta = \beta - \beta^*.$$

By Condition (v), $\bar{u}_j = 0$ for every $j \in S^*$. If $j \notin \operatorname{supp}(\beta)$, then either $j \in S^*$, in which case $\Delta_j \bar{u}_j = 0$, or $j \notin S^*$, in which case Condition (iv) gives $\Delta_j = 0$. Hence

$$\Delta_j \bar{u}_j = 0 \qquad \text{for every } j \notin \operatorname{supp}(\beta),$$

and therefore

$$\sum_{j=1}^{p} \Delta_j \bar{u}_j = \sum_{j \in \operatorname{supp}(\beta)} \Delta_j \bar{u}_j. \tag{1}$$

Also, since the support is contained in the full coordinate set and every summand is nonnegative,

$$\sum_{j\in\mathrm{supp}(\beta)} \Delta_j^2 \le \sum_{j=1}^{p} \Delta_j^2 = \|\Delta\|^2. \tag{2}$$

Set $t = \varepsilon\kappa_L$. By $\varepsilon > 0$ and Condition (vi), $t > 0$, so $t^{-1} > 0$ and

$$t^{-1}t = 1.$$

For arbitrary real numbers a, b, the elementary inequality

$$2XY \le X^2 + Y^2$$

applied to $X = ta$ and $Y = b$, followed by multiplication by the positive number t^{-1}, gives

$$2ab \le ta^2 + t^{-1}b^2 = \varepsilon\kappa_L\, a^2 + (\varepsilon\kappa_L)^{-1}b^2. \tag{3}$$

Applying (3) with $a = \Delta_j$ and $b = \bar{u}_j$, and summing over $j \in \mathrm{supp}(\beta)$, yields

$$2\sum_{j\in\mathrm{supp}(\beta)} \Delta_j\bar{u}_j \le \varepsilon\kappa_L \sum_{j\in\mathrm{supp}(\beta)} \Delta_j^2 + (\varepsilon\kappa_L)^{-1} \sum_{j\in\mathrm{supp}(\beta)} \bar{u}_j^2. \tag{4}$$

Multiplying (2) by the nonnegative factor $\varepsilon\kappa_L$, we obtain

$$\varepsilon\kappa_L \sum_{j\in\mathrm{supp}(\beta)} \Delta_j^2 \le \varepsilon\kappa_L\|\Delta\|^2 = \varepsilon\big(\kappa_L\|\Delta\|^2\big). \tag{5}$$

Condition (vii), applied to $v = \Delta$, gives

$$\kappa_L\|\Delta\|^2 \le \Delta^{\mathsf{T}}\Sigma\Delta.$$

Since $\varepsilon > 0$, this implies

$$\varepsilon\big(\kappa_L\|\Delta\|^2\big) \le \varepsilon\,\Delta^{\mathsf{T}}\Sigma\Delta. \tag{6}$$

Finally,

$$(\varepsilon\kappa_L)^{-1} = \varepsilon^{-1}\kappa_L^{-1}.$$

Combining (1), (4), (5), and (6), we get

$$\begin{aligned}
2\langle \Delta, \bar{u} \rangle &= 2 \sum_{j=1}^{p} \Delta_j \bar{u}_j \\
&= 2 \sum_{j \in \mathrm{supp}(\beta)} \Delta_j \bar{u}_j \\
&\leq \varepsilon \kappa_L \sum_{j \in \mathrm{supp}(\beta)} \Delta_j^2 + (\varepsilon \kappa_L)^{-1} \sum_{j \in \mathrm{supp}(\beta)} \bar{u}_j^2 \\
&\leq \varepsilon \kappa_L \|\Delta\|^2 + (\varepsilon \kappa_L)^{-1} \sum_{j \in \mathrm{supp}(\beta)} \bar{u}_j^2 \\
&= \varepsilon \big(\kappa_L \|\Delta\|^2 \big) + (\varepsilon \kappa_L)^{-1} \sum_{j \in \mathrm{supp}(\beta)} \bar{u}_j^2 \\
&\leq \varepsilon \, \Delta^{\mathsf{T}} \Sigma \Delta + (\varepsilon \kappa_L)^{-1} \sum_{j \in \mathrm{supp}(\beta)} \bar{u}_j^2 \\
&= \varepsilon \, \Delta^{\mathsf{T}} \Sigma \Delta + \varepsilon^{-1} \kappa_L^{-1} \sum_{j \in \mathrm{supp}(\beta)} \bar{u}_j^2.
\end{aligned}$$

Since $\Delta = \beta - \beta^*$, this is exactly

$$2\langle \beta - \beta^*, \bar{u} \rangle \leq \varepsilon \, (\beta - \beta^*)^{\mathsf{T}} \Sigma (\beta - \beta^*) + \varepsilon^{-1} \kappa_L^{-1} \sum_{j \in \mathrm{supp}(\beta)} \bar{u}_j^2,$$

as required.

□

Lemma 6. *Let $p \geq 0$ be an integer, let $\mathcal{E}$ be a finite nonempty index set, and for each $e \in \mathcal{E}$ let μ_e be a measure on $\mathbb{R}^p \times \mathbb{R}$. Let $\beta^* \in \mathbb{R}^p$. For each $e \in \mathcal{E}$ and each finite subset $S \subseteq \{1, \ldots, p\}$, let $\beta_{e,S} \in \mathbb{R}^p$. Let $\kappa > 0$. Write $z = (x, y)$ and*

$$r_{\beta^*}(z) = y - \beta^* \cdot x,$$

where $\cdot$ is the Euclidean dot product. For $j \in \{1, \ldots, p\}$, set

$$u_j(\mu, \beta^*) = \sum_{e \in \mathcal{E}} \int_{\mathbb{R}^p \times \mathbb{R}} r_{\beta^*}(x, y) \, x_j \, d\mu_e(x, y).$$

Define

$$G(\mu, \beta^*) = \{ j \in \{1, \ldots, p\} : u_j(\mu, \beta^*) \neq 0 \}.$$

For $S \subseteq \{1, \dots, p\}$, define

$$B(S) = \sum_{j \in S} u_j(\mu, \beta^*)^2, \qquad \bar{\beta}_S = \frac{1}{|\mathcal{E}|} \sum_{e \in \mathcal{E}} \beta_{e,S},$$

and

$$D(S) = \sum_{e \in \mathcal{E}} \frac{1}{|\mathcal{E}|} \|\beta_{e,S} - \bar{\beta}_S\|^2.$$

Finally let

$$\gamma_* = \kappa^{-3} \sup \left\{ \frac{B(S)}{D(S)} : S \subseteq \{1, \dots, p\},\ S \cap G(\mu, \beta^*) \neq \varnothing \right\},$$

with the conventions $\sup \varnothing = 0$ *and* $B(S)/0 = 0$ *in the displayed ratios. Assume:*

(i) each μ_e is a probability measure on $\mathbb{R}^p \times \mathbb{R}$;

(ii) $\kappa > 0$;

(iii) for every $S \subseteq \{1, \dots, p\}$ with $S \cap G(\mu, \beta^*) \neq \varnothing$, there exist $e, e' \in \mathcal{E}$ such that $\beta_{e,S} \neq \beta_{e',S}$.

Then

$$0 \leq \gamma_*$$

and, for every $S \subseteq \{1, \dots, p\}$,

$$B(S) \leq \kappa^3 \gamma_* \, D(S).$$

Proof. For each $S \subseteq \{1, \dots, p\}$, the quantity $B(S)$ is nonnegative because it is a finite sum of squares:

$$B(S) = \sum_{j \in S} u_j(\mu, \beta^*)^2 \geq 0.$$

Also $D(S) \geq 0$, since $|\mathcal{E}|^{-1} \geq 0$ and each squared norm $\|\beta_{e,S} - \bar{\beta}_S\|^2$ is nonnegative.

Let

$$\mathcal{R} = \left\{ \frac{B(S)}{D(S)} : S \subseteq \{1, \dots, p\},\ S \cap G(\mu, \beta^*) \neq \varnothing \right\}.$$

Every element of $\mathcal{R}$ is nonnegative: if the denominator is positive this follows from $B(S) \geq 0$ and $D(S) \geq 0$, and if the denominator is zero the totalized quotient is zero. Therefore

$$\sup \mathcal{R} \geq 0.$$

Since $\kappa^{-3} \geq 0$, we obtain

$$\gamma_* = \kappa^{-3} \sup \mathcal{R} \geq 0.$$

This proves the first assertion.

Now fix $S \subseteq \{1, \dots, p\}$. We first record that

$$\kappa^3 \gamma_* = \sup \mathcal{R}.$$

Indeed, substituting the definition of γ_*,

$$\kappa^3 \gamma_* = \kappa^3 \big(\kappa^{-3} \sup \mathcal{R}\big) = (\kappa^3 \kappa^{-3}) \sup \mathcal{R} = \sup \mathcal{R},$$

because $\kappa \neq 0$ by Condition (ii).

The set $\mathcal{R}$ is bounded above. The family of finite subsets of $\{1, \dots, p\}$ is finite, so the set of all numbers $B(T)/D(T)$ for such T is the image of a finite set under a real-valued map. Hence it is finite and therefore bounded above.

We split into two cases.

If $S \cap G(\mu, \beta^*) \neq \varnothing$, we claim that $D(S) > 0$. Suppose instead that $D(S) = 0$. Since $D(S)$ is a sum of nonnegative terms and equals zero, each summand is zero:

$$\frac{1}{|\mathcal{E}|} \|\beta_{e,S} - \bar{\beta}_S\|^2 = 0 \qquad \text{for every } e \in \mathcal{E}.$$

Because $|\mathcal{E}| > 0$, this forces

$$\|\beta_{e,S} - \bar{\beta}_S\|^2 = 0,$$

hence $\beta_{e,S} = \bar{\beta}_S$ for every e. Thus all $\beta_{e,S}$ are equal, contradicting Condition (iii), which gives two environments e, e' with $\beta_{e,S} \neq \beta_{e',S}$. Therefore $D(S) > 0$.

Since $S \cap G(\mu, \beta^*) \neq \varnothing$, the quotient $B(S)/D(S)$ belongs to $\mathcal{R}$. As $\sup \mathcal{R}$ is an upper bound for $\mathcal{R}$,

$$\frac{B(S)}{D(S)} \leq \sup \mathcal{R}.$$

Multiplying by the positive number $D(S)$ gives

$$B(S) \leq D(S) \sup \mathcal{R} = \kappa^3 \gamma_* \, D(S).$$

This is the desired inequality in this case.

If $S \cap G(\mu, \beta^*) = \varnothing$, then for every $j \in S$ we have $j \notin G(\mu, \beta^*)$, hence

$$u_j(\mu, \beta^*) = 0.$$

Equivalently, the defining integral vanishes; swapping the two factors inside the integral gives

the same value, so the corresponding summand in $B(S)$ vanishes. Therefore

$$B(S) = \sum_{j \in S} u_j(\mu, \beta^*)^2 = 0.$$

On the other hand, $\kappa^3 > 0$, $\gamma_* \geq 0$, and $D(S) \geq 0$, so

$$\kappa^3 \gamma_* \, D(S) \geq 0.$$

Thus

$$B(S) = 0 \leq \kappa^3 \gamma_* \, D(S).$$

This completes the proof in all cases.

□

Lemma 7 (Pooled risk expansion around a reference parameter)**.** *Let I be a finite nonempty index set, let $m = |I|$, and let $p \in \mathbb{N}$. For each $e \in I$, let μ_e be a probability measure on $\mathbb{R}^p \times \mathbb{R}$. Let $\beta, \beta^* \in \mathbb{R}^p$, and let $\omega_e = m^{-1}$ for every $e \in I$. For $z = (x, y) \in \mathbb{R}^p \times \mathbb{R}$, write*

$$\varepsilon_{\beta^*}(z) = y - (\beta^*)^\top x.$$

Define, for $e \in I$,

$$R_e(\beta) = \int_{\mathbb{R}^p \times \mathbb{R}} \left(y - \beta^\top x\right)^2 d\mu_e(x, y),$$

and let Σ_e be the $p \times p$ matrix with entries

$$(\Sigma_e)_{ij} = \int_{\mathbb{R}^p \times \mathbb{R}} x_i x_j \, d\mu_e(x, y).$$

For a $p \times p$ matrix $M = (M_{ij})$ and a vector $v = (v_1, \ldots, v_p)$, set

$$Q(M, v) = \sum_{i=1}^{p} \sum_{j=1}^{p} M_{ij} v_i v_j.$$

Let

$$R_{\text{pool}}(\beta) = \sum_{e \in I} \omega_e R_e(\beta), \qquad \Sigma_{\text{pool}} = \frac{1}{m} \sum_{e \in I} \Sigma_e.$$

Assume the following integrability conditions:

(i) for every $e \in I$, the function $z \mapsto \varepsilon_{\beta^*}(z)^2$ is μ_e-integrable;

(ii) for every $e \in I$, the function $z = (x, y) \mapsto (y - \beta^\top x)^2$ is μ_e-integrable;

(iii) for every $e \in I$ and every $j \in \{1, \dots, p\}$, the function $z \mapsto \varepsilon_{\beta^*}(z)x_j$ is μ_e-integrable;

(iv) for every $e \in I$ and every $i, j \in \{1, \dots, p\}$, the function $z = (x, y) \mapsto x_i x_j$ is μ_e-integrable.

Then

$$R_{\text{pool}}(\beta) - R_{\text{pool}}(\beta^*) = Q\big(\Sigma_{\text{pool}}, \beta - \beta^*\big) - 2\sum_{j=1}^{p}(\beta_j - \beta_j^*)\left(\sum_{e \in I}\omega_e \int_{\mathbb{R}^p \times \mathbb{R}} \varepsilon_{\beta^*}(z)\, x_j \, d\mu_e(z)\right).$$

Proof. Put $\Delta = \beta - \beta^*$. We first prove the corresponding identity for a single environment $e \in I$. For $z = (x, y)$, define

$$a(z) = y - (\beta^*)^\top x = \varepsilon_{\beta^*}(z), \qquad b(z) = \beta^\top x - (\beta^*)^\top x = \Delta^\top x.$$

Then $y - \beta^\top x = a(z) - b(z)$, and pointwise

$$(a(z) - b(z))^2 - a(z)^2 = b(z)^2 - 2a(z)b(z).$$

The integrability assumptions make the following functions μ_e-integrable. By Condition (i), a^2 is μ_e-integrable; by Condition (ii), $(a - b)^2$ is μ_e-integrable. Since

$$b(z)^2 = \sum_{i=1}^{p}\sum_{j=1}^{p} \Delta_i \Delta_j \, x_i x_j,$$

Condition (iv) and finite summation with constant coefficients show that b^2 is μ_e-integrable. Similarly, since

$$a(z)b(z) = \sum_{j=1}^{p} \Delta_j \, \varepsilon_{\beta^*}(z) x_j,$$

Condition (iii) shows that ab is μ_e-integrable.

Using linearity of the integral with respect to the subtraction of the μ_e-integrable functions $(a - b)^2$ and a^2, we get

$$R_e(\beta) - R_e(\beta^*) = \int \big((a - b)^2 - a^2\big)\, d\mu_e = \int \big(b^2 - 2ab\big)\, d\mu_e.$$

Since b^2 and ab are μ_e-integrable, this becomes

$$R_e(\beta) - R_e(\beta^*) = \int b^2 \, d\mu_e - 2\int ab \, d\mu_e.$$

Now expanding $b(z) = \sum_{j=1}^{p} \Delta_j x_j$ and using finite linearity of the integral gives

$$\int b^2 \, d\mu_e = \sum_{i=1}^{p} \sum_{j=1}^{p} \Delta_i \Delta_j \int x_i x_j \, d\mu_e = Q(\Sigma_e, \Delta).$$

Likewise,

$$\int ab \, d\mu_e = \sum_{j=1}^{p} \Delta_j \int \varepsilon_{\beta^*}(z) x_j \, d\mu_e(z).$$

Therefore, for each $e \in I$,

$$R_e(\beta) - R_e(\beta^*) = Q(\Sigma_e, \Delta) - 2 \sum_{j=1}^{p} \Delta_j \int \varepsilon_{\beta^*}(z) x_j \, d\mu_e(z).$$

Next we pool these identities over $e \in I$. By the definition of the pooled risk and finite linearity of summation,

$$R_{\text{pool}}(\beta) - R_{\text{pool}}(\beta^*) = \sum_{e \in I} \omega_e \big(R_e(\beta) - R_e(\beta^*) \big).$$

Substituting the single-environment identity yields

$$R_{\text{pool}}(\beta) - R_{\text{pool}}(\beta^*) = \sum_{e \in I} \omega_e Q(\Sigma_e, \Delta) - 2 \sum_{e \in I} \omega_e \sum_{j=1}^{p} \Delta_j \int \varepsilon_{\beta^*}(z) x_j \, d\mu_e(z).$$

We next rewrite the first term. From the definition of Q,

$$Q\Big(\sum_{e \in I} M_e, v \Big) = \sum_{e \in I} Q(M_e, v), \qquad Q(cM, v) = cQ(M, v),$$

for every finite family of matrices M_e, every vector v, and every scalar c; these identities follow by interchanging the finite sums in the definition of Q. Since $\omega_e = m^{-1}$,

$$\sum_{e \in I} \omega_e Q(\Sigma_e, \Delta) = \frac{1}{m} \sum_{e \in I} Q(\Sigma_e, \Delta) = Q\left(\frac{1}{m} \sum_{e \in I} \Sigma_e, \Delta \right) = Q(\Sigma_{\text{pool}}, \Delta).$$

For the second term, interchanging the finite sums over e and j gives

$$\sum_{e \in I} \omega_e \sum_{j=1}^{p} \Delta_j \int \varepsilon_{\beta^*}(z) x_j \, d\mu_e(z) = \sum_{j=1}^{p} \Delta_j \left(\sum_{e \in I} \omega_e \int \varepsilon_{\beta^*}(z) x_j \, d\mu_e(z) \right).$$

Combining the two computations and recalling that $\Delta_j = \beta_j - \beta_j^*$, we obtain

$$R_{\mathrm{pool}}(\beta) - R_{\mathrm{pool}}(\beta^*) = Q\big(\Sigma_{\mathrm{pool}}, \beta - \beta^*\big) - 2\sum_{j=1}^{p}(\beta_j - \beta_j^*)\left(\sum_{e\in I}\omega_e \int \varepsilon_{\beta^*}(z)x_j\, d\mu_e(z)\right).$$

This is the desired identity.

□

Main Theorem

Theorem 8 (Global strong convexity with respect to $\beta^\star$)**.** *Let $p \in \mathbb{N}$, let $\mathcal{E}$ be a nonempty finite index set, and for each $e \in \mathcal{E}$ let μ_e be a measure on $\mathbb{R}^p \times \mathbb{R}$. Write a generic point of $\mathbb{R}^p \times \mathbb{R}$ as (x, y), with $x = (x_1, \dots, x_p)$. For $\beta \in \mathbb{R}^p$, put*

$$\operatorname{supp}\beta = \{j \in \{1, \dots, p\} : \beta_j \neq 0\}, \qquad \beta^T x = \sum_{j=1}^{p}\beta_j x_j,$$

and define the population mean squared prediction error in environment e by

$$\mathcal{R}_e(\beta) = \int_{\mathbb{R}^p\times\mathbb{R}} (y - \beta^T x)^2\, d\mu_e(x, y).$$

Let $\beta^\star \in \mathbb{R}^p$, $S^\star \subseteq \{1, \dots, p\}$, and $\sigma^2 > 0$. Let

$$\mathcal{F}_{S^\star} = \sigma\big(x_j : j \in S^\star\big)$$

be the σ-algebra generated by the coordinates indexed by $S^\star$.

Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space, let $n \geq 1$, and for each $e \in \mathcal{E}$ and $i = 1, \dots, n$ let

$$(X_{e,i}, Y_{e,i}) : \Omega \to \mathbb{R}^p \times \mathbb{R}$$

be random variables. Let $\omega_e \in \mathbb{R}$ for $e \in \mathcal{E}$, let $\beta^{(e,S)} \in \mathbb{R}^p$ for $e \in \mathcal{E}$ and $S \subseteq \{1, \dots, p\}$, and let $\kappa_L, \kappa_U \in \mathbb{R}$.

Define the second-moment matrices

$$\Sigma_e = (\Sigma_{e,jk})_{1\leq j,k\leq p}, \qquad \Sigma_{e,jk} = \int_{\mathbb{R}^p\times\mathbb{R}} x_j x_k\, d\mu_e(x, y),$$

and, for vectors $v \in \mathbb{R}^p$,

$$v^T \Sigma_e v = \sum_{j=1}^{p} \sum_{k=1}^{p} \Sigma_{e,jk} v_j v_k, \qquad \bar{\Sigma} = \frac{1}{|\mathcal{E}|} \sum_{e \in \mathcal{E}} \Sigma_e.$$

Put

$$u_j^{(e)} = \int_{\mathbb{R}^p \times \mathbb{R}} x_j \big(y - \beta^{\star T} x\big)\, d\mu_e(x,y), \qquad \bar{u}_j = \frac{1}{|\mathcal{E}|} \sum_{e \in \mathcal{E}} u_j^{(e)},$$

and define the spurious-coordinate set

$$G = \left\{ j : \sum_{e \in \mathcal{E}} u_j^{(e)} \neq 0 \right\}.$$

For $S \subseteq \{1, \dots, p\}$, *set*

$$b_S = \sum_{j \in S} \bar{u}_j^2, \qquad \bar{\beta}^{(S)} = \frac{1}{|\mathcal{E}|} \sum_{e \in \mathcal{E}} \beta^{(e,S)}, \qquad d_S = \sum_{e \in \mathcal{E}} \frac{1}{|\mathcal{E}|} \big\| \beta^{(e,S)} - \bar{\beta}^{(S)} \big\|^2,$$

and define

$$\gamma^\star = \kappa_L^{-3} \sup \left\{ \frac{b_S}{d_S} : S \subseteq \{1, \dots, p\},\ S \cap G \neq \varnothing \right\},$$

with the convention that a quotient with zero denominator is 0. *Define the regularization functional*

$$J(\beta) = \sum_{j \in \operatorname{supp} \beta} \left(\sum_{e \in \mathcal{E}} \omega_e \int_{\mathbb{R}^p \times \mathbb{R}} x_j (y - \beta^T x)\, d\mu_e(x,y) \right)^2$$

and the objective

$$Q_\gamma(\beta) = \sum_{e \in \mathcal{E}} \omega_e\, \mathcal{R}_e(\beta) + \gamma J(\beta).$$

Assume the following conditions.

(i) $S^\star = \operatorname{supp} \beta^\star$, *and* $\sigma^2 > 0$.

(ii) For every $e \in \mathcal{E}$, μ_e *is a probability measure; for every* $j = 1, \dots, p$, $x_j \in L^2(\mu_e)$ *and*

$$\int_{\mathbb{R}^p \times \mathbb{R}} x_j^2\, d\mu_e(x,y) \leq \sigma^2;$$

moreover, μ_e*-almost everywhere,*

$$\mathbb{E}_{\mu_e}[y \mid \mathcal{F}_{S^\star}](x,y) = \beta^{\star T} x,$$

and

$$\mathbb{E}_{\mu_e}\left[(y - \mathbb{E}_{\mu_e}[y \mid \mathcal{F}_{S^\star}])^2 \mid \mathcal{F}_{S^\star}\right](x,y) \le \sigma^2.$$

(iii) *For every $e \in \mathcal{E}$, $y \in L^2(\mu_e)$.*

(iv) *For every $e \in \mathcal{E}$, the vector field x is μ_e-integrable and*

$$\int_{\mathbb{R}^p \times \mathbb{R}} x \, d\mu_e(x,y) = 0.$$

(v) *For every $e \in \mathcal{E}$ and $i = 1, \dots, n$, the random variable $(X_{e,i}, Y_{e,i})$ is measurable, its law under $\mathbb{P}$ is μ_e, and the finite family*

$$\{(X_{e,i}, Y_{e,i}) : e \in \mathcal{E},\ i = 1, \dots, n\}$$

is jointly independent under $\mathbb{P}$.

(vi) *For every $e \in \mathcal{E}$,*

$$\omega_e > 0, \qquad \sum_{e \in \mathcal{E}} \omega_e = 1, \qquad \omega_e = \frac{1}{|\mathcal{E}|}.$$

(vii) *For every $e \in \mathcal{E}$ and every $S \subseteq \{1, \dots, p\}$,*

$$\operatorname{supp} \beta^{(e,S)} \subseteq S,$$

and $\beta^{(e,S)}$ is a global minimizer of $\mathcal{R}_e$ on the set $\{\beta \in \mathbb{R}^p : \operatorname{supp} \beta \subseteq S\}$.

(viii) *$0 < \kappa_L \le 1$, $1 \le \kappa_U$, and for every $e \in \mathcal{E}$ and every $v \in \mathbb{R}^p$,*

$$\kappa_L \|v\|^2 \le v^T \Sigma_e v \le \kappa_U \|v\|^2.$$

(ix) *For every $S \subseteq \{1, \dots, p\}$ with $S \cap G \neq \varnothing$, there exist $e, e' \in \mathcal{E}$ such that*

$$\beta^{(e,S)} \neq \beta^{(e',S)}.$$

Then the following two assertions hold:

$$\forall\, \varepsilon \in (0,1),\ \forall\, \gamma \in \mathbb{R} \text{ with } \gamma \ge \varepsilon^{-1}\gamma^\star,\ \forall\, \beta \in \mathbb{R}^p,$$

$$Q_\gamma(\beta) - Q_\gamma(\beta^\star) \ge (1-\varepsilon)\,(\beta - \beta^\star)^T \bar{\Sigma} (\beta - \beta^\star) + \kappa_L^2 \left(\gamma - \varepsilon^{-1}\gamma^\star\right) d_{\operatorname{supp}\beta},$$

and

$$\forall\, \gamma > \gamma^\star,\ \forall\, \beta \in \mathbb{R}^p, \quad \beta \neq \beta^\star \implies Q_\gamma(\beta^\star) < Q_\gamma(\beta).$$

Proof. We work throughout under the assumptions of the theorem. Write a generic point as $z = (x, y) \in \mathbb{R}^p \times \mathbb{R}$. Since $\mathcal{E}$ is a nonempty finite set, its cardinality satisfies $|\mathcal{E}| > 0$. The σ-algebra $\mathcal{F}_{S^\star}$ generated by the coordinates x_j with $j \in S^\star$ is a sub-σ-algebra of the Borel σ-algebra on $\mathbb{R}^p \times \mathbb{R}$, so the conditional expectations below are well defined with respect to the measures μ_e.

For each $e \in \mathcal{E}$, let

$$m_e(x, y) = \mathbb{E}_{\mu_e}[y \mid \mathcal{F}_{S^\star}](x, y).$$

By Condition (ii), μ_e is a probability measure, $m_e(x, y) = \beta^{\star T} x$ for μ_e-almost every (x, y), and

$$\mathbb{E}_{\mu_e}\big[(y - m_e)^2 \mid \mathcal{F}_{S^\star}\big] \leq \sigma^2 \qquad \mu_e\text{-a.e.}$$

Applying the tower property of conditional expectation gives

$$\int (y - m_e)^2 \, d\mu_e = \int \mathbb{E}_{\mu_e}\big[(y - m_e)^2 \mid \mathcal{F}_{S^\star}\big] \, d\mu_e \leq \int \sigma^2 \, d\mu_e = \sigma^2.$$

This is the first variance bound: the mean squared deviation of y from its $\mathcal{F}_{S^\star}$-conditional mean is at most σ^2.

Since $y - \beta^{\star T} x = y - m_e$ μ_e-a.e., the conditional expectation of $(y - \beta^{\star T} x)^2$ equals that of $(y - m_e)^2$ μ_e-a.e. Hence

$$\mathbb{E}_{\mu_e}\big[(y - \beta^{\star T} x)^2 \mid \mathcal{F}_{S^\star}\big] \leq \sigma^2 \qquad \mu_e\text{-a.e.},$$

and, integrating,

$$\int (y - \beta^{\star T} x)^2 \, d\mu_e = \int (y - m_e)^2 \, d\mu_e \leq \sigma^2.$$

Thus the residual $y - \beta^{\star T} x$ has second moment at most σ^2 in every environment e.

Next we record the L^2 facts for the coordinates. By Condition (ii), $x_j \in L^2(\mu_e)$ and $\int x_j^2 \, d\mu_e \leq \sigma^2$ for every j. Hence each coordinate function x_j is square-integrable. By Hölder's inequality, the product $x_i x_j$ belongs to $L^1(\mu_e)$ for every i, j. Consequently, for any $b \in \mathbb{R}^p$,

$$(b^T x)^2 = \sum_{i,j} b_i b_j \, x_i x_j$$

is a finite linear combination of integrable functions, so $(b^T x)^2 \in L^1(\mu_e)$. Also $b^T x = \sum_j b_j x_j$ is measurable as a finite linear combination of the coordinate functions; hence $b^T x \in L^2(\mu_e)$ whenever $(b^T x)^2$ is integrable.

We now compare the integrability of y^2 with membership of y in L^2. Because y is measurable, for each e,

$$y^2 \in L^1(\mu_e) \quad \Longleftrightarrow \quad y \in L^2(\mu_e).$$

Thus the assertion that y^2 is integrable for every e is equivalent to the assertion that $y \in L^2(\mu_e)$ for every e.

Several implications follow. First, if y^2 is μ_e-integrable for every e, then $y \in L^2(\mu_e)$. Since $\beta^{\star T} x \in L^2(\mu_e)$, the residual $y - \beta^{\star T} x$ is a difference of two L^2-functions, hence belongs to $L^2(\mu_e)$. Therefore its square $(y - \beta^{\star T} x)^2$ is μ_e-integrable for every e. Second, if $y \in L^2(\mu_e)$ for every e, then trivially y^2 is μ_e-integrable, and the same difference argument gives $(y - \beta^{\star T} x)^2 \in L^1(\mu_e)$. Conversely, if $(y - \beta^{\star T} x)^2$ is μ_e-integrable for every e, then $y - \beta^{\star T} x \in L^2(\mu_e)$; adding the L^2-function $\beta^{\star T} x$ gives $y \in L^2(\mu_e)$, hence $y^2 \in L^1(\mu_e)$.

We also need the degenerate case in which y^2 fails to be integrable. Suppose for some e that $y^2 \notin L^1(\mu_e)$. For any $b \in \mathbb{R}^p$, if $(y - b^T x)^2$ were μ_e-integrable, then $y - b^T x \in L^2(\mu_e)$. Since $b^T x \in L^2(\mu_e)$, adding these two L^2-functions would give $y \in L^2(\mu_e)$, hence $y^2 \in L^1(\mu_e)$, a contradiction. Therefore $(y - b^T x)^2$ is not μ_e-integrable. In the real-valued integral convention used for the population mean squared prediction error functional, a nonintegrable integrand contributes the value 0; hence

$$\mathcal{R}_e(b) = \int (y - b^T x)^2 \, d\mu_e = 0$$

for every b.

If this nonintegrability holds for every $e \in \mathcal{E}$, then for every b we have $\mathcal{R}_e(b) = 0$ for all e, so the weighted pooled mean squared prediction error vanishes:

$$\sum_{e \in \mathcal{E}} \omega_e \, \mathcal{R}_e(b) = 0.$$

Moreover, for each e and each coordinate j, the function

$$t \longmapsto \mathcal{R}_e(b + t e_j)$$

is identically zero, where e_j is the j-th standard basis vector. Its derivative at $t = 0$ is therefore zero. Since this derivative is, up to the factor -2, the coordinate integral $\int x_j (y - b^T x) \, d\mu_e$, those coordinate integrals vanish. Consequently every summand in the regularizer $J(b)$, which is a sum of squares of such weighted coordinate integrals over the coordinates in $\operatorname{supp} b$, is zero. Hence

$$J(b) = 0.$$

Therefore

$$Q_\gamma(b) = \sum_{e\in\mathcal{E}} \omega_e \, \mathcal{R}_e(b) + \gamma J(b) = 0$$

for every $\gamma \in \mathbb{R}$ and every $b \in \mathbb{R}^p$.

In particular, in this degenerate regime both $Q_\gamma(\beta^\star)$ and $Q_\gamma(b)$ equal 0 for every γ and every b. Thus the strict inequality

$$Q_\gamma(\beta^\star) < Q_\gamma(b)$$

is impossible, since it would read $0 < 0$.

It follows that if the strict comparison property

$$\forall\, \gamma > \gamma^\star,\ \forall\, \beta \in \mathbb{R}^p, \quad \beta \neq \beta^\star \Longrightarrow Q_\gamma(\beta^\star) < Q_\gamma(\beta)$$

holds, while the nonintegrability $y^2 \notin L^1(\mu_e)$ holds for every e, then necessarily $\beta = \beta^\star$ for every β. Indeed, if some $\beta \neq \beta^\star$ existed, taking $\gamma = \gamma^\star + 1 > \gamma^\star$ would give $Q_\gamma(\beta^\star) < Q_\gamma(\beta)$, contradicting the fact that both values are zero.

We now return to the nondegenerate situation ensured by Condition (iii). By Condition (iii), $y \in L^2(\mu_e)$ for every e. Hence y^2 is μ_e-integrable for every e. Applying the implication above, the residual square $(y - \beta^{\star T}x)^2$ is μ_e-integrable for every e. Since $y - \beta^{\star T}x$ is measurable, it follows that

$$y - \beta^{\star T}x \in L^2(\mu_e)$$

for every e. Adding the L^2-function $\beta^{\star T}x$ gives again $y \in L^2(\mu_e)$, and therefore $y^2 \in L^1(\mu_e)$ for every e.

We also obtain a quantitative bound. For every $z = (x, y)$,

$$y^2 \leq 2(y - \beta^{\star T}x)^2 + 2(\beta^{\star T}x)^2.$$

Indeed, writing $y = (y - \beta^{\star T}x) + \beta^{\star T}x$ and using $2ab \leq a^2 + b^2$, we get

$$y^2 = (y - \beta^{\star T}x)^2 + 2(y - \beta^{\star T}x)(\beta^{\star T}x) + (\beta^{\star T}x)^2 \leq 2(y - \beta^{\star T}x)^2 + 2(\beta^{\star T}x)^2.$$

If y^2 is μ_e-integrable, integrating this pointwise inequality gives

$$\int y^2 \, d\mu_e \leq 2\int (y - \beta^{\star T}x)^2 \, d\mu_e + 2\int (\beta^{\star T}x)^2 \, d\mu_e \leq 2\sigma^2 + 2\int (\beta^{\star T}x)^2 \, d\mu_e,$$

where we used the residual second-moment bound $\int (y - \beta^{\star T}x)^2 \, d\mu_e \leq \sigma^2$. If y^2 is not μ_e-

integrable, then the left-hand integral is 0 under the real-valued integral convention, while the right-hand side is nonnegative because $\sigma^2 > 0$ and the integral of the square $(\beta^{\star T} x)^2$ is nonnegative. Thus the same inequality

$$\int y^2 \, d\mu_e \le 2\sigma^2 + 2 \int (\beta^{\star T} x)^2 \, d\mu_e$$

holds for every e in all cases.

Finally, since y is measurable and y^2 is μ_e-integrable, we have $y \in L^2(\mu_e)$ for every e. Because μ_e is a probability measure, $L^2(\mu_e) \subseteq L^1(\mu_e)$. Hence y is μ_e-integrable for every e, that is,

$$\int |y| \, d\mu_e < \infty \qquad \text{for all } e \in \mathcal{E}.$$

These are the L^2- and L^1-facts for the response that are needed in the subsequent estimates.

We now build the analytic and first-order machinery needed for the quadratic lower bound, and then combine it with the comparison through $\gamma^\star$ to obtain the two assertions of the theorem.

Integrability of the cross terms. Throughout, points of $\mathbb{R}^p \times \mathbb{R}$ are written $z = (x, y)$ with $x = (x_1, \ldots, x_p)$, and all integrals below are taken with respect to the fixed environment measure μ_e.

First, for each $e \in \mathcal{E}$ and each coordinate j, the function $z \mapsto y\, x_j$ is μ_e-integrable. Indeed $y \in L^2(\mu_e)$ by Condition (iii) and $x_j \in L^2(\mu_e)$ by Condition (ii), so their product lies in $L^1(\mu_e)$ by Hölder's inequality.

Second, for each e and each fixed vector $b \in \mathbb{R}^p$, the function $z \mapsto y\,(b^T x)$ is μ_e-integrable: since $b^T x = \sum_j b_j x_j$, we have $y(b^T x) = \sum_j b_j\,(y x_j)$, a finite linear combination of the integrable functions just considered.

Third, for each e, b, and j, the function $z \mapsto (y - b^T x) x_j$ is μ_e-integrable. Writing

$$(y - b^T x) x_j = y x_j - (b^T x) x_j,$$

the first summand is integrable by the first point, while

$$(b^T x) x_j = \sum_{i=1}^{p} b_i \, (x_i x_j)$$

is a finite linear combination of products $x_i x_j$; each product is μ_e-integrable because $x_i, x_j \in L^2(\mu_e)$ (Condition (ii)) implies $x_i x_j \in L^1(\mu_e)$, and finite sums of integrable functions are integrable. Hence the difference is integrable.

Fourth, for each e and b, the squared residual $z \mapsto (y - b^T x)^2$ is μ_e-integrable. Expanding

$$(y - b^T x)^2 = y^2 - 2y(b^T x) + (b^T x)^2,$$

the term y^2 is integrable since $y \in L^2(\mu_e)$, the term $2y(b^T x)$ is integrable by the second point, and

$$(b^T x)^2 = \sum_{i,j} b_i b_j \, (x_i x_j)$$

is a finite linear combination of the integrable products $x_i x_j$. Their sum is therefore integrable.

These four facts ensure that every population mean squared prediction error $\mathcal{R}_e(\beta)$ is finite and that the products appearing in its expansion are genuine integrable functions, so that the differentiations and expansions below are legitimate.

The first-order differentiation formula. For each e, b, and j, denote by $\partial_j \mathcal{R}_e(b)$ the j-th partial derivative of the map $\beta \mapsto \mathcal{R}_e(\beta)$ at $\beta = b$, that is, the derivative at $t = 0$ of the one-variable function $t \mapsto \mathcal{R}_e(b + te_j)$. Applying Lemma 1 with the integrability of $(y - b^T x)^2$, the integrability of $(y - b^T x)x_j$, and the vanishing first moment $\int x_j \, d\mu_e = 0$ supplied by Condition (iv), we obtain

$$\partial_j \mathcal{R}_e(b) = -2 \int_{\mathbb{R}^p \times \mathbb{R}} (y - b^T x) x_j \, d\mu_e(x, y).$$

Thus the gradient of the population mean squared prediction error is expressed through the first moments of the residual times the coordinates.

Orthogonality at the true parameter. We apply Lemma 3, using the identity $S^\star = \operatorname{supp} \beta^\star$ from Condition (i), the true-law conditions from Condition (ii) together with the vanishing first moment of Condition (iv), the square-integrability $y \in L^2$ of Condition (iii), and the residual-coordinate integrability just established. Its conclusion has two parts. First, for every $e \in \mathcal{E}$ and every $j \in S^\star$,

$$u_j^{(e)} = \int_{\mathbb{R}^p \times \mathbb{R}} x_j \, (y - \beta^{\star T} x) \, d\mu_e(x, y) = 0.$$

Indeed $y - \beta^{\star T} x$ has conditional mean zero given the σ-algebra $\mathcal{F}_{S^\star}$ of Condition (ii), and for $j \in S^\star$ the coordinate x_j is $\mathcal{F}_{S^\star}$-measurable, so pulling out the measurable factor makes

the expectation vanish. Second, averaging over the environments, for every $j \in S^\star$,

$$\bar{u}_j = \frac{1}{|\mathcal{E}|} \sum_{e \in \mathcal{E}} u_j^{(e)} = 0.$$

The coefficient identity. Using the vanishing of $\bar{u}_j$ on $S^\star$, the coefficient

$$b_{\operatorname{supp}\beta} = \sum_{j \in \operatorname{supp}\beta} \bar{u}_j^2$$

equals the sum over the enlarged index set $\operatorname{supp}\beta \cup S^\star$: enlarging the summation set can only add terms $\bar{u}_j^2$ with $j \in S^\star \setminus \operatorname{supp}\beta$, and all of these are zero. Hence

$$b_{\operatorname{supp}\beta} = \sum_{j \in \operatorname{supp}\beta} \bar{u}_j^2 = \sum_{j \in \operatorname{supp}\beta} \Big(\frac{1}{|\mathcal{E}|} \sum_{e \in \mathcal{E}} u_j^{(e)} \Big)^2.$$

Vanishing of the regularizer at the true parameter. We claim $J(\beta^\star) = 0$. For any coordinate $j \in \operatorname{supp}\beta^\star$ we have $j \in S^\star$ by Condition (i). By the differentiation formula and the orthogonality just proved,

$$\partial_j \mathcal{R}_e(\beta^\star) = -2 \int_{\mathbb{R}^p \times \mathbb{R}} x_j \, (y - \beta^{\star T} x) \, d\mu_e(x, y) = 0$$

for every e. Since each summand of $J(\beta^\star)$ is a squared first moment of this form and vanishes, summing over $j \in \operatorname{supp}\beta^\star$ and over e gives $J(\beta^\star) = 0$.

Algebra of the quadratic form. For any scalar c, any matrix M, and any vector v,

$$v^T (cM) v = c \, (v^T M v),$$

as is immediate from the definition $v^T M v = \sum_{i,j} M_{ij} v_i v_j$. Likewise, for any finite family of matrices $(M_e)_{e \in \mathcal{E}}$ and any v,

$$v^T \Big(\sum_{e \in \mathcal{E}} M_e \Big) v = \sum_{e \in \mathcal{E}} v^T M_e v,$$

by linearity of the finite sum in the entries.

The pooled Loewner bound. For every $v \in \mathbb{R}^p$,

$$v^T \bar{\Sigma}\, v = \frac{1}{|\mathcal{E}|} \sum_{e \in \mathcal{E}} v^T \Sigma_e v,$$

using the previous two algebraic facts and the definition $\bar{\Sigma} = |\mathcal{E}|^{-1} \sum_e \Sigma_e$. By Condition (viii), $v^T \Sigma_e v \geq \kappa_L \|v\|^2$ for every e, so summing gives

$$\sum_{e \in \mathcal{E}} v^T \Sigma_e v \geq |\mathcal{E}|\, \kappa_L \|v\|^2.$$

Multiplying by $|\mathcal{E}|^{-1} > 0$ yields

$$v^T \bar{\Sigma}\, v \geq \kappa_L \|v\|^2 \qquad \text{for all } v \in \mathbb{R}^p.$$

In particular, since $\kappa_L > 0$, the quadratic form of $\bar{\Sigma}$ is nonnegative: $v^T \bar{\Sigma}\, v \geq 0$ for all v.

Nonnegativity of the coefficient and heterogeneity quantities. For every $S \subseteq \{1, \ldots, p\}$, the quantity $b_S = \sum_{j \in S} \bar{u}_j^2$ is a sum of squares, hence $b_S \geq 0$. Likewise

$$d_S = \sum_{e \in \mathcal{E}} \frac{1}{|\mathcal{E}|} \left\| \beta^{(e,S)} - \bar{\beta}^{(S)} \right\|^2 \geq 0,$$

each term being a nonnegative scalar times a squared norm.

The degenerate case. Suppose now that y is not square-integrable in any environment, i.e., $z \mapsto y^2$ fails to be μ_e-integrable for every e. As established earlier, this degenerate situation forces every population mean squared prediction error to vanish, and consequently the objective

$$Q_\gamma(\beta) = 0 \qquad \text{for all } \gamma \text{ and all } \beta.$$

We record that in this situation the asserted quadratic lower bound cannot hold in the sharp form. Fix $\varepsilon \in (0, 1)$, a real γ with $\gamma \geq \varepsilon^{-1} \gamma^\star$, and a vector $\beta \neq \beta^\star$. The left-hand side of the putative inequality is

$$(1 - \varepsilon)(\beta - \beta^\star)^T \bar{\Sigma} (\beta - \beta^\star) + \kappa_L^2 \big(\gamma - \varepsilon^{-1} \gamma^\star\big) d_{\operatorname{supp} \beta}.$$

Because $\beta \neq \beta^\star$, we have $\beta - \beta^\star \neq 0$, hence $\|\beta - \beta^\star\| > 0$ and, by the pooled Loewner bound, $(\beta - \beta^\star)^T \bar{\Sigma} (\beta - \beta^\star) \geq \kappa_L \|\beta - \beta^\star\|^2 > 0$; multiplying by $1 - \varepsilon > 0$ keeps this first term strictly positive. The second term is nonnegative, since $\gamma - \varepsilon^{-1} \gamma^\star \geq 0$, $\kappa_L^2 \geq 0$, and $d_{\operatorname{supp} \beta} \geq 0$.

Hence the whole left-hand side is strictly positive. But $Q_\gamma(\beta) - Q_\gamma(\beta^\star) = 0 - 0 = 0$, and a strictly positive number cannot be ≤ 0. Therefore the inequality

$$(1-\varepsilon)(\beta-\beta^\star)^T\bar{\Sigma}(\beta-\beta^\star) + \kappa_L^2(\gamma - \varepsilon^{-1}\gamma^\star)d_{\operatorname{supp}\beta} \le Q_\gamma(\beta) - Q_\gamma(\beta^\star)$$

fails. This confirms that square-integrability of the response is indispensable for the strong-convexity statement.

The main quadratic lower bound. We now prove that for every $\varepsilon > 0$, every real γ with $\varepsilon^{-1}\gamma^\star \le \gamma$, and every β,

$$Q_\gamma(\beta) - Q_\gamma(\beta^\star) \ge (1-\varepsilon)(\beta-\beta^\star)^T\bar{\Sigma}(\beta-\beta^\star) + \kappa_L^2\big(\gamma - \varepsilon^{-1}\gamma^\star\big)d_{\operatorname{supp}\beta}.$$

First record the properties of $\gamma^\star$. By Lemma 6, applied with the probability measures, the constrained minimizers $\beta^{(e,S)}$, and the constant $\kappa_L > 0$, we have

$$\gamma^\star \ge 0, \qquad \text{and} \qquad b_S \le \kappa_L^3\,\gamma^\star\, d_S \quad \text{for every } S \subseteq \{1,\dots,p\}.$$

The second inequality expresses that each ratio b_S/d_S is dominated by the supremum defining $\gamma^\star$ (and is trivial when $d_S = 0$, where the ratio is interpreted as 0).

Fix $\varepsilon > 0$, γ, and β with $\varepsilon^{-1}\gamma^\star \le \gamma$. Since $\varepsilon^{-1}\gamma^\star \ge 0$ and $\gamma \ge \varepsilon^{-1}\gamma^\star$, we have $\gamma \ge 0$.

Next, the displacement $\Delta := \beta - \beta^\star$ vanishes outside $\operatorname{supp}\beta \cup S^\star$. Indeed, if $j \notin \operatorname{supp}\beta \cup S^\star$, then $\beta_j = 0$ by the definition of the support, and $\beta_j^\star = 0$ because $j \notin S^\star = \operatorname{supp}\beta^\star$ (Condition (i)); hence $\Delta_j = \beta_j - \beta_j^\star = 0$.

Because $J(\beta^\star) = 0$, the difference of objectives is

$$Q_\gamma(\beta) - Q_\gamma(\beta^\star) = \Big(\sum_{e\in\mathcal{E}} \omega_e \mathcal{R}_e(\beta) - \sum_{e\in\mathcal{E}} \omega_e \mathcal{R}_e(\beta^\star)\Big) + \gamma J(\beta).$$

For the difference of the population mean squared prediction errors we use Lemma 7. Writing $c = y - \beta^{\star T}x$ and $d = (\beta - \beta^\star)^T x$, so that $y - \beta^T x = c - d$, we have

$$(y-\beta^T x)^2 - (y-\beta^{\star T}x)^2 = d^2 - 2cd.$$

Integrating, $\int d^2 = \Delta^T\Sigma_e\Delta$ and $\int c\,d = \sum_j \Delta_j u_j^{(e)}$; therefore, using the uniform weights $\omega_e = |\mathcal{E}|^{-1}$,

$$\sum_{e\in\mathcal{E}} \omega_e \mathcal{R}_e(\beta) - \sum_{e\in\mathcal{E}} \omega_e \mathcal{R}_e(\beta^\star) = \Delta^T\bar{\Sigma}\,\Delta - 2\sum_{j=1}^p \Delta_j\,\bar{u}_j.$$

The needed integrability is exactly what was verified above: the squared residuals at β and at $\beta^\star$, the residual-coordinate products at $\beta^\star$, and the coordinate products $x_i x_j$ are all integrable.

We note that the averaged first moments are weighted uniformly: the vector with coordinates $\sum_e \omega_e u_j^{(e)}$ coincides, because $\omega_e = |\mathcal{E}|^{-1}$, with the vector $\bar{u} = (\bar{u}_j)$ whose coordinates are $\bar{u}_j = |\mathcal{E}|^{-1} \sum_e u_j^{(e)}$; and $\bar{u}_j = 0$ for $j \in S^\star$ by the orthogonality established above.

For the linear term we use Lemma 5, applied with the support condition $S^\star = \operatorname{supp} \beta^\star$, the vanishing $\bar{u}_j = 0$ on $S^\star$, the constant $\kappa_L > 0$, and the pooled Loewner bound. Since Δ vanishes off $\operatorname{supp} \beta \cup S^\star$ and $\bar{u}$ vanishes on $S^\star$, we have $\sum_j \Delta_j \bar{u}_j = \sum_{j \in \operatorname{supp} \beta} \Delta_j \bar{u}_j$, and the lemma gives

$$2 \sum_{j=1}^{p} \Delta_j \, \bar{u}_j \leq \varepsilon \, \Delta^T \bar{\Sigma} \, \Delta + \varepsilon^{-1} \kappa_L^{-1} \, b_{\operatorname{supp} \beta}.$$

This is the chain obtained from the CauchySchwarz bound $2|\Delta^T \bar{u}| \leq 2\|\Delta\| \, \|\bar{u}_{\operatorname{supp} \beta}\|$, the elementary inequality $2xy \leq tx^2 + t^{-1}y^2$ with $t = \varepsilon \kappa_L > 0$, and the estimate $\kappa_L \|\Delta\|^2 \leq \Delta^T \bar{\Sigma} \Delta$. Substituting into the expansion of the population mean squared prediction error,

$$\sum_e \omega_e \mathcal{R}_e(\beta) - \sum_e \omega_e \mathcal{R}_e(\beta^\star) \geq (1 - \varepsilon) \, \Delta^T \bar{\Sigma} \, \Delta - \varepsilon^{-1} \kappa_L^{-1} \, b_{\operatorname{supp} \beta}.$$

For the regularizer we use Lemma 2, which gives

$$J(\beta) \geq \kappa_L^2 \, d_{\operatorname{supp} \beta}.$$

Since $\gamma \geq 0$, multiplying by γ yields

$$\gamma J(\beta) \geq \gamma \, \kappa_L^2 \, d_{\operatorname{supp} \beta}.$$

Finally, we control the negative coefficient term using the $\gamma^\star$ bound. Since $b_{\operatorname{supp} \beta} \leq \kappa_L^3 \gamma^\star d_{\operatorname{supp} \beta}$ and $-\varepsilon^{-1} \kappa_L^{-1} \leq 0$, multiplying the inequality by this nonpositive factor reverses it:

$$-\varepsilon^{-1} \kappa_L^{-1} b_{\operatorname{supp} \beta} \geq -\varepsilon^{-1} \kappa_L^{-1} \, \kappa_L^3 \, \gamma^\star \, d_{\operatorname{supp} \beta} = -\varepsilon^{-1} \kappa_L^2 \, \gamma^\star \, d_{\operatorname{supp} \beta},$$

where we used $\kappa_L^{-1} \kappa_L^3 = \kappa_L^2$, valid because $\kappa_L \neq 0$.

Combining the three estimates,

$$Q_\gamma(\beta) - Q_\gamma(\beta^\star) \geq (1-\varepsilon) \, \Delta^T \bar{\Sigma} \, \Delta - \varepsilon^{-1} \kappa_L^{-1} \, b_{\operatorname{supp} \beta} + \gamma \, \kappa_L^2 \, d_{\operatorname{supp} \beta} \geq (1-\varepsilon) \, \Delta^T \bar{\Sigma} \, \Delta + \kappa_L^2 \big(\gamma - \varepsilon^{-1} \gamma^\star\big) d_{\operatorname{supp} \beta},$$

which is the asserted lower bound.

Strict comparison. We prove that for every $\gamma > \gamma^\star$ and every $\beta \neq \beta^\star$, $Q_\gamma(\beta^\star) < Q_\gamma(\beta)$.

Fix such γ and β. Since $\beta \neq \beta^\star$, the displacement $\Delta = \beta - \beta^\star$ is nonzero, so $\|\Delta\| > 0$ and the pooled Loewner bound gives

$$\Delta^T \bar{\Sigma}\, \Delta > 0.$$

We produce $\varepsilon \in (0,1)$ with $\varepsilon^{-1}\gamma^\star \leq \gamma$. If $\gamma^\star > 0$, then $\gamma > \gamma^\star > 0$ and $0 < \gamma^\star/\gamma < 1$; set $\varepsilon = (1 + \gamma^\star/\gamma)/2$. Then $\varepsilon > 1/2 > 0$, and since $\gamma^\star/\gamma < 1$ we get $\varepsilon < 1$; moreover $\varepsilon \geq \gamma^\star/\gamma$ (as $1 \geq \gamma^\star/\gamma$), so multiplying by $\gamma > 0$ gives $\gamma \geq \varepsilon^{-1}\gamma^\star$. If instead $\gamma^\star \leq 0$, take $\varepsilon = 1/2$; then $0 < \varepsilon < 1$ and $\varepsilon^{-1}\gamma^\star = 2\gamma^\star \leq \gamma^\star < \gamma$, since $\gamma^\star \leq 0 < \gamma$.

With this ε the main lower bound applies (as $\varepsilon > 0$ and $\varepsilon^{-1}\gamma^\star \leq \gamma$):

$$Q_\gamma(\beta) - Q_\gamma(\beta^\star) \geq (1-\varepsilon)\, \Delta^T \bar{\Sigma}\, \Delta + \kappa_L^2 \big(\gamma - \varepsilon^{-1}\gamma^\star\big) d_{\operatorname{supp}\beta}.$$

The second term is nonnegative, because $\gamma - \varepsilon^{-1}\gamma^\star \geq 0$, $\kappa_L^2 \geq 0$, and $d_{\operatorname{supp}\beta} \geq 0$. The first term is strictly positive, because $1 - \varepsilon > 0$ and $\Delta^T \bar{\Sigma}\, \Delta > 0$. Hence

$$Q_\gamma(\beta) - Q_\gamma(\beta^\star) > 0,$$

that is, $Q_\gamma(\beta^\star) < Q_\gamma(\beta)$. This establishes the strict comparison for all $\gamma > \gamma^\star$ and all $\beta \neq \beta^\star$.

Both assertions of the theorem now follow: the strong-convexity lower bound for $\varepsilon \in (0,1)$ and $\gamma \geq \varepsilon^{-1}\gamma^\star$ is the main quadratic estimate (which even holds for all $\varepsilon > 0$), and the strict minimisation of $\beta^\star$ at every $\gamma > \gamma^\star$ is the strict comparison just obtained.

□

Task outcomes and dependencies

Task registry snapshot: paper_fresh_20261028t224543381z_eills, revision 28, captured 2026-10-29T02:15:40.345Z.

This report belongs to paper_fresh_20261028t224543381z_eills (job paper_fresh_20261028t224543381 its own terminal report determines its outcome, since registry settlement can lag reporting.

paper_2c08ddb98e7c4dc4993a0b23e7a07337: external; pending; dependencies: none.

External assumption linked to task paper_2c08ddb98e7c4dc4993a0b23e7a07337: retained as external. No completed proof for this consumer is certified by this snapshot.

Each task has its own outcome. A proved child does not certify the main task or discharge other external assumptions.